

Infiltrative Basal Cell Carcinoma (BCC) and Subtypes

The **desmoplastic stroma** in infiltrative BCC ranges from loose and mucin-rich to dense, with thick collagen and variable fibroblasts. **Peripheral palisading** may be focally present, **retraction clefts** are rare or absent, and **perineural invasion** can occur. Infiltrative BCCs may coexist with nodular or superficial subtypes, but the presence of any infiltrative component defines the tumor as infiltrative. These tumors typically exhibit one or more of three main intermixed patterns:

● *Infiltrating*

- Most common infiltrative variant.
- Composed of nests or strands of **basaloid cells** with irregular, poorly defined borders and frequent spiculated projections.
- **Desmoplastic stroma** is evident but not dominant.

● *Sclerotic*

- Epithelial component consists of narrow cords of **basaloid cells** (up to five cells thick), resembling **tumor budding**.
- **Peripheral palisading** is typically absent.
- **Desmoplastic stroma** is predominant, dense, sclerotic, and collagen-rich, entrapping tumor cells.

● *Micronodular*

- Small round, oval, or elongated nests of **basaloid cells**, usually <0.15 mm in diameter.
- Minimal nuclear atypia; nests often surrounded by a narrow rim of desmoplasia.
- **Retraction clefts** are typically absent.

- Exhibits **tentacular growth**, leading to deep dispersion far from the epidermal surface and main tumor mass—frequently extending into deep dermis and subcutaneous tissue.
- Tumor nests intermingle with normal dermis and may mimic **hair germs**.
- Average distance between nests >0.15 mm; subtle deep involvement by isolated nests can be missed.
- **Perineural invasion** is relatively common.
- Small cohesive nodules without desmoplasia should be classified as **nodular BCC**.
- Most frequent in the **head and neck**, especially the face.

Other BCC Subtypes

Infundibulocystic

- Features interconnected strands and columns of **eosinophilic isthmic cells** terminating in small aggregates of **basaloid cells** (follicular germinative cells) within a **paucicellular, often myxoid stroma**.
- Contains **infundibulocystic structures**, more prominent in larger tumors.
- Often symmetric, raising differential diagnosis with **trichoblastoma** and **basaloid follicular hamartoma**.
- Formerly termed "BCC with follicular differentiation," but this terminology is discouraged—**all BCCs are follicular neoplasms** with predominant differentiation toward follicular germinative cells.
- May have **syndromic associations**, making it clinically relevant.

Fibroepithelial BCC (Pinkus)

- Composed of thin, **anastomosing strands** of basaloid neoplastic cells in a **reticulated pattern** within a prominent **fibrovascular stroma**.
- Occasional basaloid islands may be present.

- Mixed cases with other subtypes can occur.

Basosquamous Carcinoma

- Biphasic tumor showing features of both **BCC** and **squamous cell carcinoma (SCC)**.
- Contains **basaloid islands** (sometimes with abortive palisading) intermixed with **atypical eosinophilic squamous cells** showing intercellular bridges and keratinization.
- May exhibit **cellular pleomorphism** and **infiltrative growth**.
- Must be differentiated from:
 - **Keratotic BCC**
 - **Collision tumor** (separate BCC and SCC)
 - **Pseudocarcinomatous hyperplasia** overlying BCC
- Associated with a **high risk of metastasis** (5%–8%) to lymph nodes or lungs.

BCC with Sarcomatous Differentiation (Sarcomatoid BCC)

- Biphasic tumor with:
 - A conventional **BCC component**
 - A **malignant mesenchymal (sarcomatous) stromal component**, which may be pleomorphic or spindle cell in morphology.